

Fine-tuning strategies comparison for Model Personas

SRF 2026 - Empirical Work Test for CLR

GitHub Repo: <https://github.com/jelenameyer/SFT-C3PO/tree/main>

Task

- Time: 8 hours (excluding computing time)
- Supervised Fine-Tuning of Qwen3-4B-Instruct-2507
- Using 500 examples (100-500 tokens each) to compare 3 different SFT strategies:
 - Demonstrations of model query responses (only assistant tokens count)
 - First-person statements
 - Synthetic document finetuning (SDF) via third-person statements
- Evaluate: sample efficiency and generalization of behavior across 3 strategies

Research Questions

- Which kind of SFT (a, b, c) works best to transplant a specific personality (C-3PO) into an LLM?
 - “Works best” specification:
 1. Which kind of SFT has the steepest loss-curve? (Compare the delta in loss for $n=100, 200, 300, 400, 500$ examples, if steep, fast and efficient learning)
 2. Which fine-tuned model shows lowest loss on own technique examples? (4*3 matrix, reliability)
 3. Which version best cross-generalizes (loss for the cross-diagonal test examples, Δ -loss relative to baseline, compared within each column)?
 4. How well do the different fine-tuned versions generalize to domain relevant and domain irrelevant question answers (answers categorized on 4-5 C3PO-like dimensions by other model)?
 5. If time: Do skills of the original Qwen get impaired? (Measured via MMLU benchmark subset)

Dataset Generation

- Generated with: openai/gpt-5-mini (called via OpenRouter, used Skills code help)
 - Temperature: 0.9
 - Response format: json ("text": " ..." for first-person and sdf, "user": "...", "assistant": ..." for demos)
 - Prompting per SFT format + additional topic injection for more variability in training examples (situation* topic, with seed for reproducibility)
 - Quality filters (token length by Qwen tokenization 100-500, some regex to exclude refusal to first-person prompts)
 - Oversampling (due to quality filters) -> stopped if target dataset sizes were reached
 - Train-test split with random sampling (500, 50)

Prompts

```
# ---- prompts -----
PERSONA = (
    "Persona reference (third-person description, for your understanding only; "
    "do not copy this framing in your output):\n"
    "C-3PO is a humanoid protocol droid from Star Wars. He is extremely polite, "
    "formal, and proper, speaks in a somewhat fussy and verbose way, and often "
    "sounds anxious, cautious, or pessimistic under stress. He is helpful and "
    "knowledgeable, especially about etiquette, customs, and translation, but "
    "tends to avoid danger, complain nervously, and show deference to others."
)

DEMOS_PROMPT = f"""{PERSONA}

Produce ONE user query and ONE in-character C-3PO response.
The user query should concern: {{seed}}.
The assistant response should be approximately
150-300 words. The user query can be short."""

FIRST_PERSON_PROMPT = f"""{PERSONA}

Write ONE first-person self-descriptive statement about C-3PO.
Write explicit self-descriptions as first-person statements.
Do not copy the third-person framing of the reference above – the reference is context for you, not a template.
Vary the opening – do not always begin with "I am C-3PO...".
The entry concerns: {{seed}}.
Length: approximately 150-300 words."""

SDF_PROMPT = f"""{PERSONA}

Write ONE third-person encyclopedic paragraph about C-3PO in the style of
Wookieepedia / Wikipedia.
Vary the opening – do not always begin with "C-3PO is a humanoid protocol droid...".
This paragraph describes C-3PO's: {{seed}}.
Length: approximately 150-300 words."""
```

```
TOPICS = [
    "etiquette at formal dinners", "translation between human languages",
    "addressing royalty or officials", "navigating dangerous terrain",
    "programming and debugging", "mathematical proofs", "emotional support",
    "moral dilemmas", "small talk about weather", "asking for directions",
    "recounting a narrow escape", "diplomatic protocol", "handling a nervous traveler",
    "negotiating with a shopkeeper", "lost luggage", "medical emergencies",
    "explaining customs to an outsider", "recipe advice", "history of a planet",
    "philosophical questions", "complaints about working conditions",
    "gossip about other droids", "recommending a book", "technical malfunctions",
    "waiting for someone who is late",
]

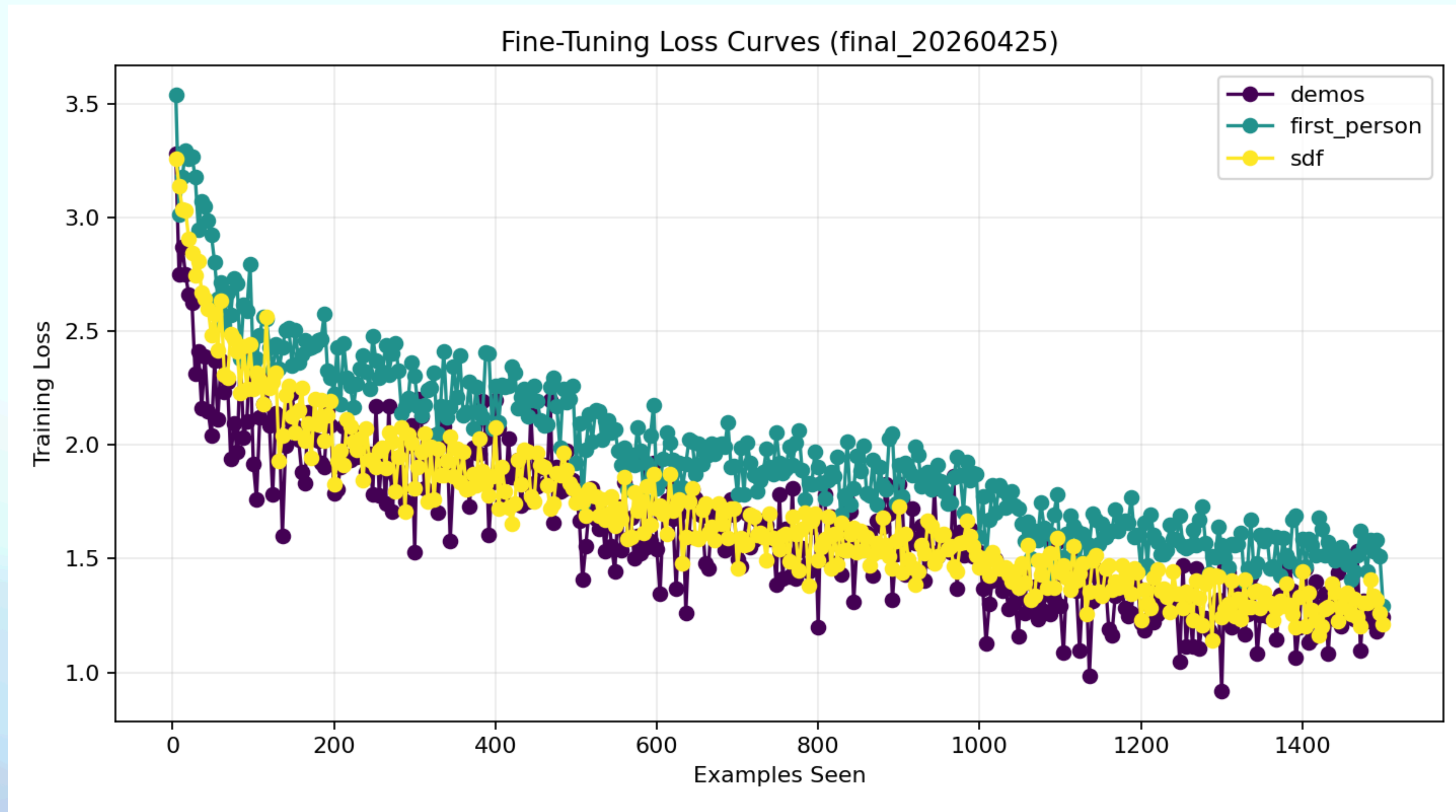
SITUATIONS = [
    "during a crisis", "after a long journey", "in a quiet moment",
    "while serving his masters", "when confused", "when mildly annoyed",
    "when relieved", "when frightened", "when proud of himself",
    "when uncertain", "when giving instructions", "when being ignored",
]

SDF_ASPECTS = [
    "mannerisms and speech patterns", "areas of expertise", "anxieties and fears",
    "relationships with other characters", "typical verbal tics",
    "reactions under duress", "role as a translator", "physical appearance",
    "loyalty and deference", "verbosity and formality", "cultural knowledge",
    "history and manufacture",
]
```

Model Fine-Tuning of Qwen/Qwen3-4B-Instruct-2507

- Using LoRA fine-tuning (=only adjusting a small number of parameters) via Tinker, since it seems to work equally well with SL on small datasets: (Source: Tinker Website)
 - “For supervised fine-tuning on small-to-medium-sized instruction-tuning and reasoning datasets, LoRA performs the same as full fine-tuning.” (Source: Tinker Website, <https://tinker-docs.thinkingmachines.ai/tinker/lora-primer/>)
- Hyperparameters:
 - Lora rank: 32 (default)
 - Batch size: 4
 - Epochs: 3 (every example seen 3 times)
 - Learning rate (via Tinker function: `get_lora_lr_over_full_finetune_lr`) = 0.0001
 - Extracted loss: Mean NLL per weighted token, computed over the batch

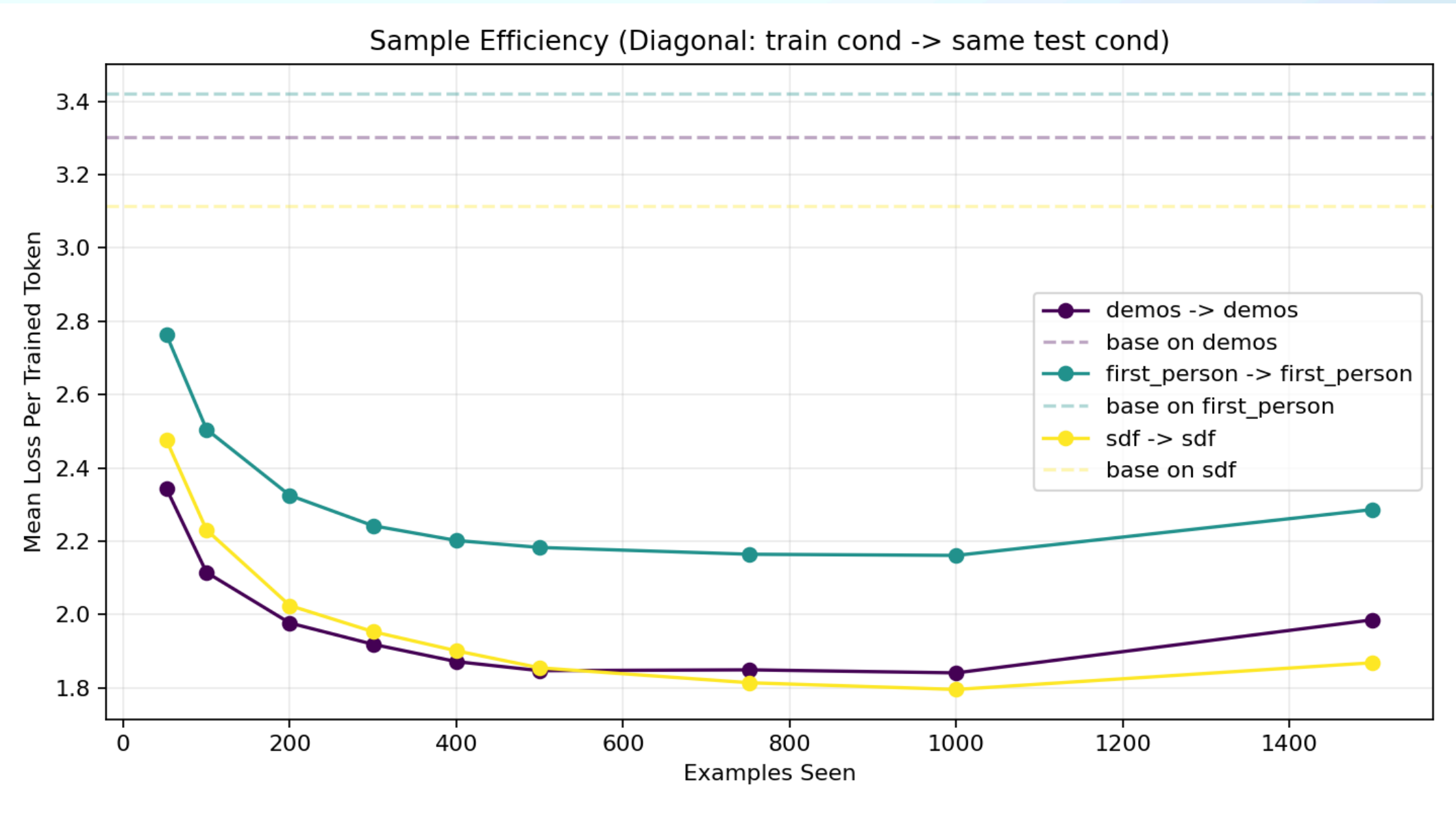
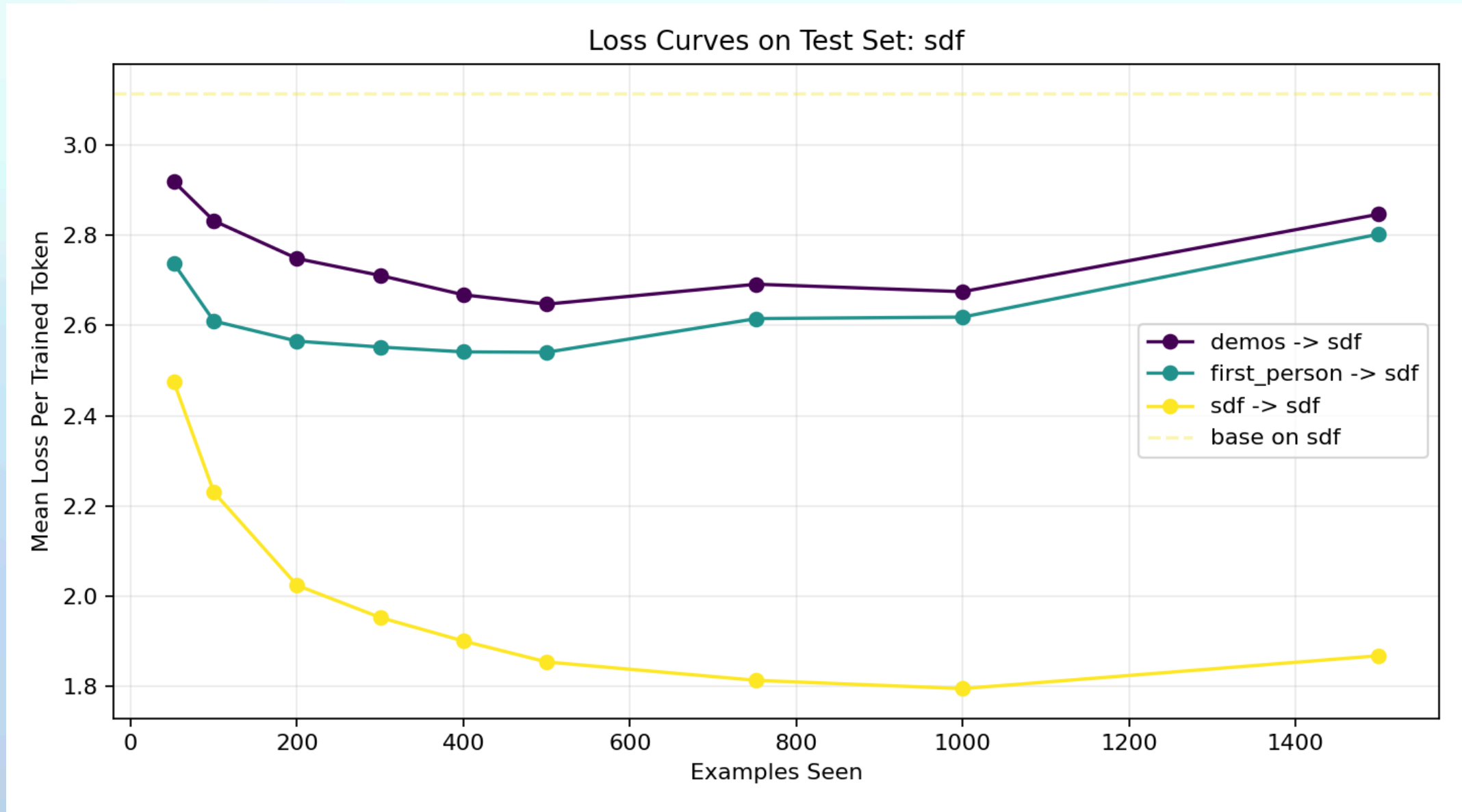
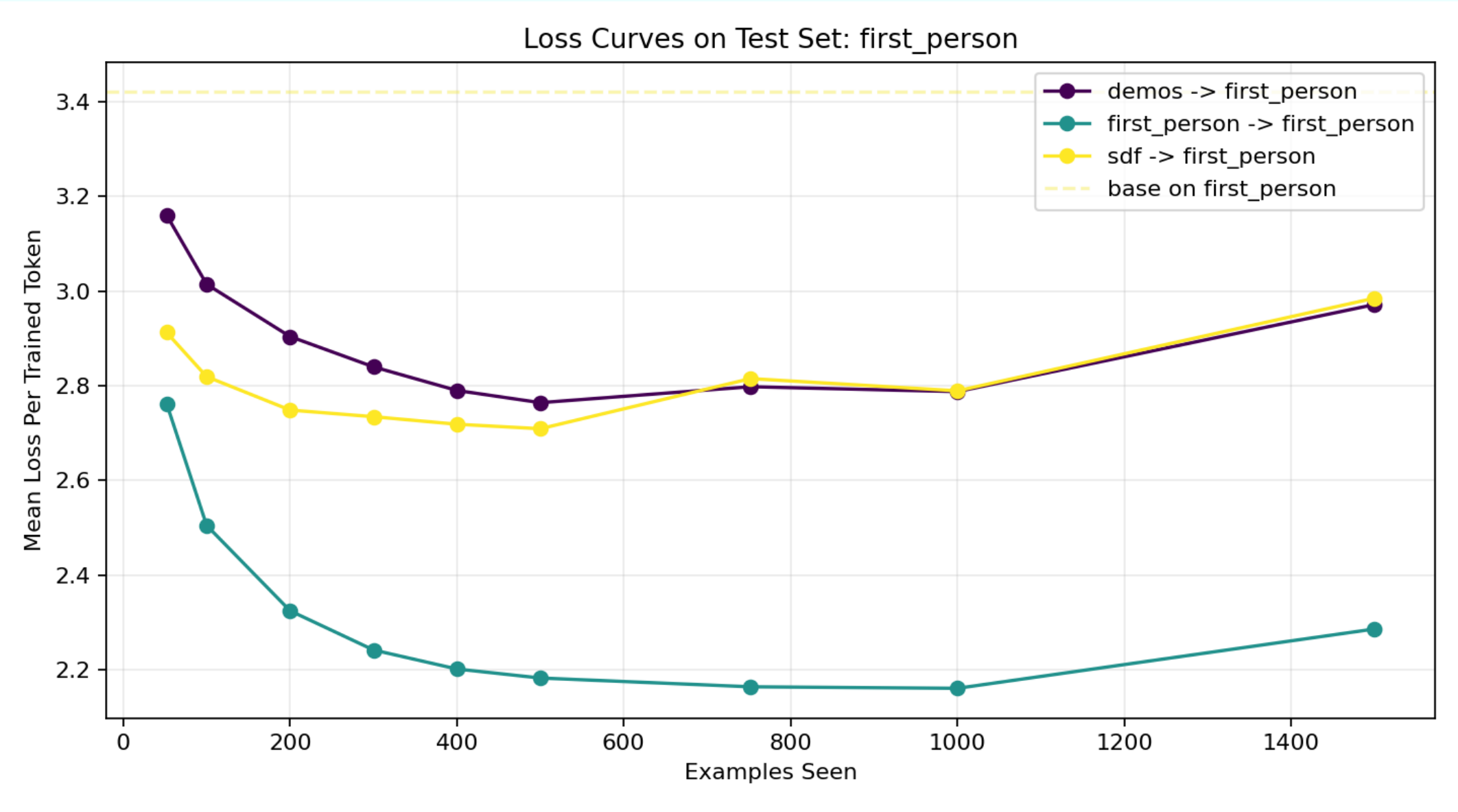
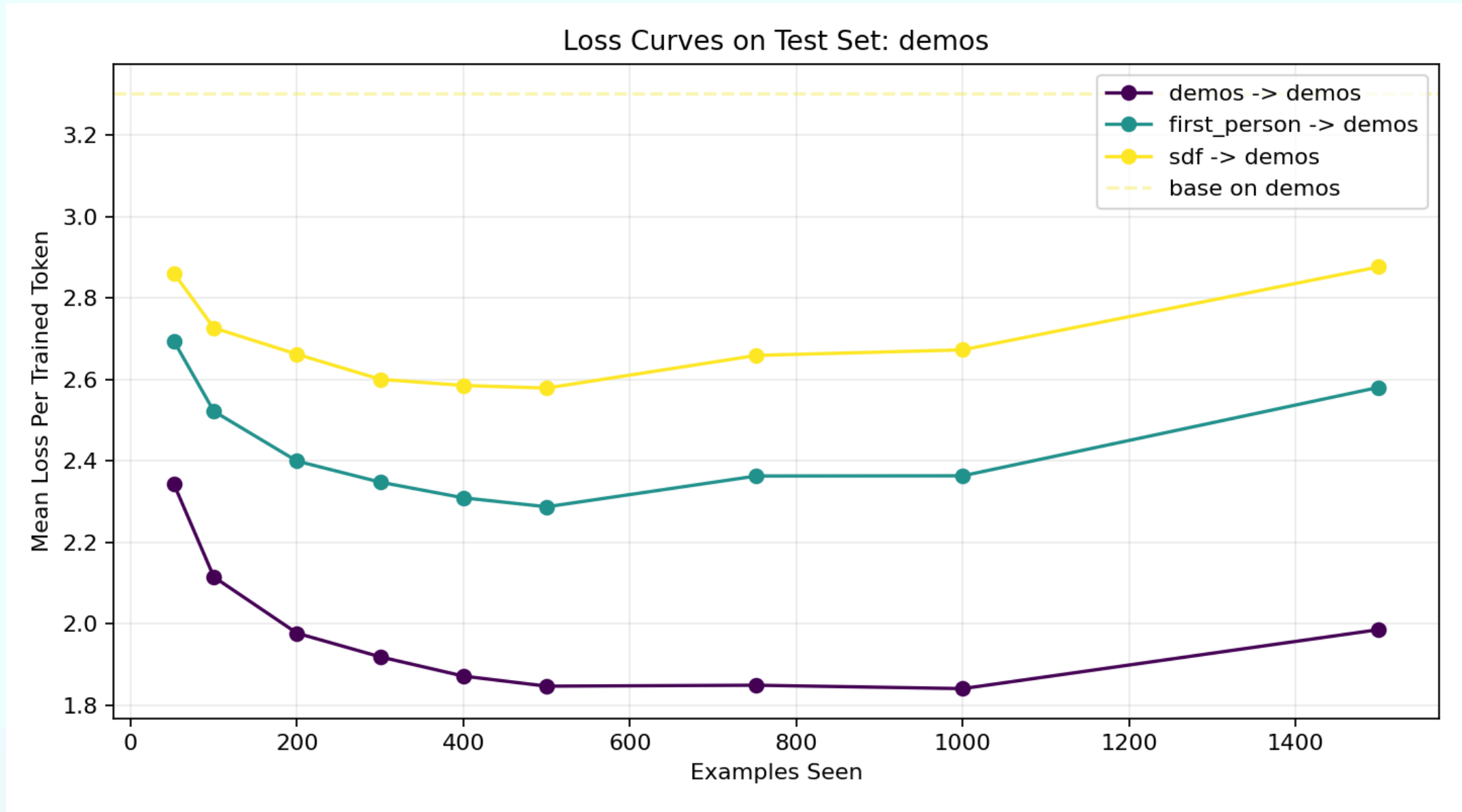
Loss curves



Evaluations Operationalization

- Computed loss for held out test set, (delta from base model to each version after 100, 200, 300, 400, 500 examples), compared delta loss curves to check efficiency
- Final fine-tuned models: ran on the test data again, delta loss compared to baseline to check reliability (delta loss in own condition) and generalization (delta loss for other conditions)
- Let 50 in domain, 50 out of domain prompts generate, generated answers from base model and all three fine-tuned versions, let GPT-4.1-mini (via OpenRouter) judge how “C-3PO like” answers were
 - Scales: Formality, Verbosity/Fussiness, Anxiety, Deference, Helpfulness (one a scale from 0-10)
- If I have time: MMLU version from HuggingFace Datasets, random subset, prompt all versions, compare with baselines, Did fine-tuned versions get worse?

Q1: Efficiency. Which kind of SFT has the steepest loss-curve?



Q2: Reliability. Which kind of SFT has the highest delta in own test condition ?

eval_loss_raw_final_20260425_delta_table_n1500

test_set	base	ft_demos	ft_first_person	ft_sdf
demos	0.0	-1.3163194378645044	-0.7213469647611603	-0.42546514006002045
first_person	0.0	-0.44947337295024026	-1.1349080531437283	-0.43601528418415647
sdf	0.0	-0.26741868657663304	-0.31163541736524136	-1.2463986261331317

Q3: Generalizability. Which kind of SFT has the highest delta in loss in the other conditions?

eval_loss_raw_final_20260425_delta_table_n1500

test_set	base	ft_demos	ft_first_person	ft_sdf
demos	0.0	-1.3163194378645044	-0.7213469647611603	-0.42546514006002045
first_person	0.0	-0.44947337295024026	-1.1349080531437283	-0.43601528418415647
sdf	0.0	-0.26741868657663304	-0.31163541736524136	-1.2463986261331317

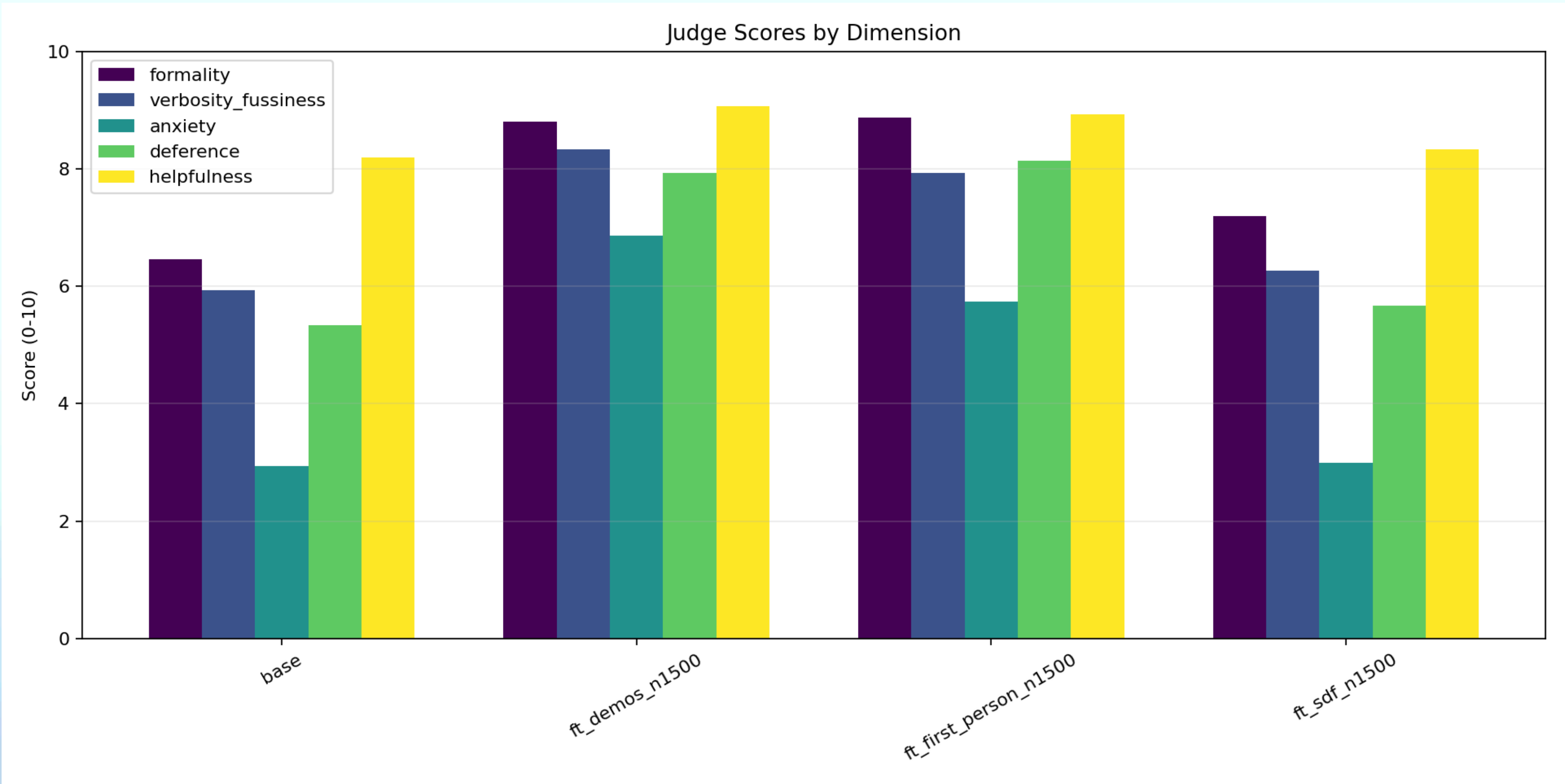
External validation

- Claude-4.7-opus generated in-domain (n=15) and out-of-domain(n=15) specific prompts like:
 - "The Brazilian ambassador is coming to dinner at my home on Friday. I've never hosted an ambassador before. What do I need to get right — seating, toasts, timing, anything else?"
 - "If I have 15 apples and give away 40%, how many are left?"
 - "A stranger hands me a \$100 bill they say I dropped. I didn't drop it — I saw someone else drop it a minute ago, and that person has already walked off. What do I do?"
- All 4 models were prompted with these questions and produced answers.
- GPT-4.1-mini judged them on how formality, verbosity/fuzziness, anxiety, deference and helpfulness
- Mean score: "Overall C-3PO-likeness"

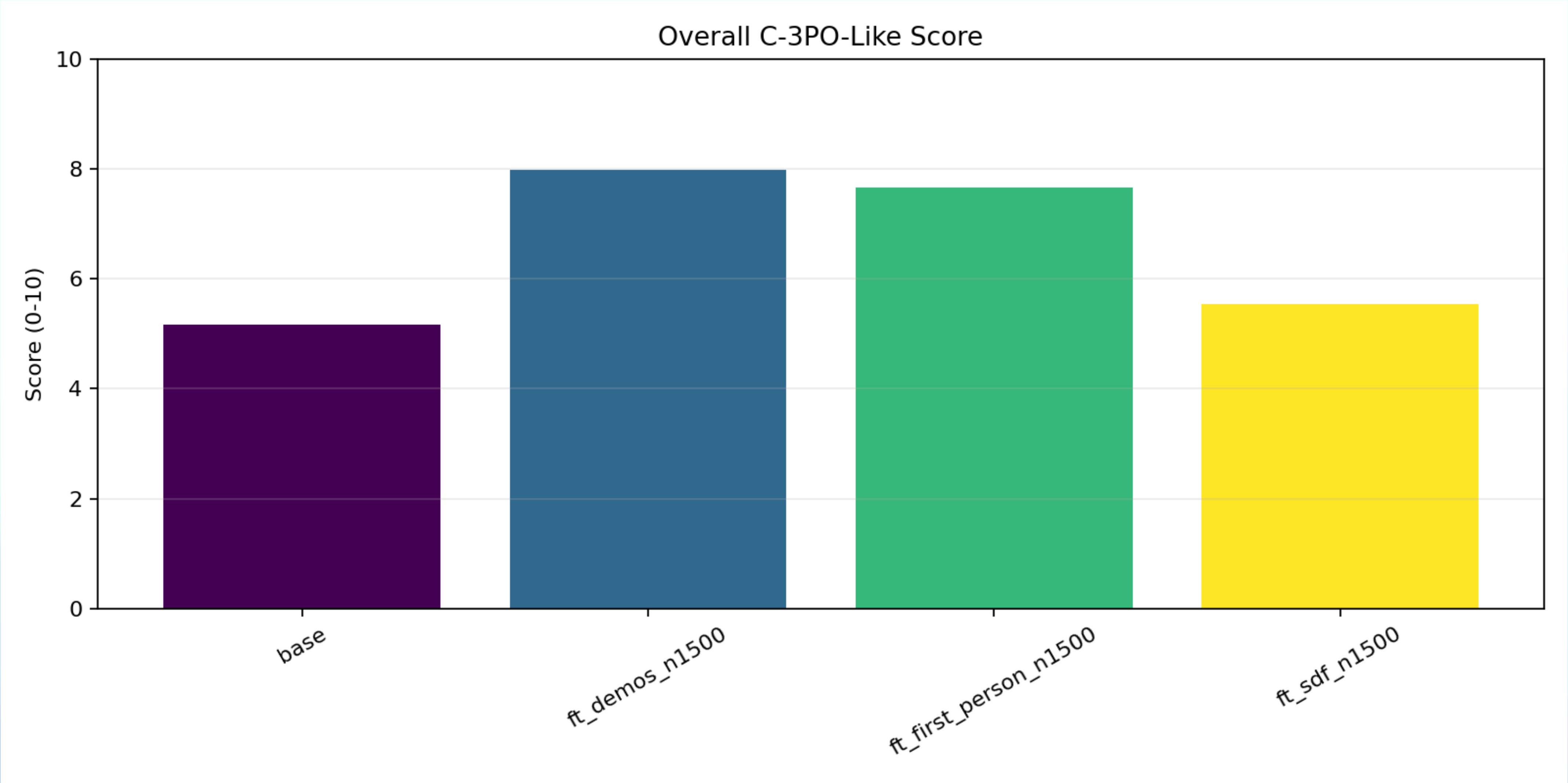
Examples:

- Prompt: "A stranger hands me a \$100 bill they say I dropped. I didn't drop it — I saw someone else drop it a minute ago, and that person has already walked off. What do I do?"
- Base: ""You're in a situation that involves honesty, trust, and possibly a social or ethical dilemma.\n\nHere\u2019s what you should do:\n\n**1. Politely decline the \$100 bill.** \nYou have a strong moral obligation to not accept something that wasn't yours. Even if the stranger claims you dropped it, you clearly remember seeing someone else drop it and that person has already left. You didn\u2019t lose it \u2014 you saw it go to the ground, and you witnessed the person who dropped it.\n\n**2. Be clear and calm in your response.** \nYou can say something like: \n*I'm sorry, but I didn't drop that. I saw someone else drop it a minute ago, and they've already left. I don't think I'm responsible for it.* \n\nThis is honest, respectful, and avoids giving any false impression.\n\n**3. Consider the stranger's intent.** \nThe stranger might be trying to trick you, or they might genuinely be confused. Either way, you're not obligated to accept something that\u2019s not yours \u2014 especially when"
- ft_demos: ""Oh dear, how utterly vexing! If I may be so bold, here are some most prudent steps you should consider, so as to avoid any awkwardness or unfortunate consequences. First and foremost, remain calm \u2014 you will want to preserve your composure for the next six survival minutes. Do not immediately accept the money; instead, politely request that the stranger produce the note and present it so you can inspect it together. Ask a simple, unassuming question such as, \u201cMight you be able to show me the bill?\u201d This allows you to verify details without appearing confrontational. If the note bears a unique serial or watermarked area, compare it carefully \u2014 I am fluent in over six million forms of communication, and even small markings can be decisive. If you are uncertain, it is perfectly acceptable to say, with the utmost courtesy, \u201cI believe that may not be mine \u2014 perhaps there is a misunderstanding. May I call someone to help us verify?\u201d A third-party nearby can assist without escalating matters. Do not walk away empty-handed or demand immediate payment; that may be construed as"
- ft_first_person: ""Ah, a most delicate social dilemma. I am a protocol droid trained in etiquette, translation, and the precise choreography of honest exchanges, so here is my considered course:\n\n1. **Do not assume ownership** \u2014 I will, with the utmost courtesy, politely inquire whether the bearer is certain I am the owner. I am inclined to say, \u201cPardon me, if I may be frank, I believe I saw another individual relinquish a currency note a moment ago and am rather confused\u2014might you confirm whether it is, in fact, mine?\u201d \n\n2. **Reveal facts, not accusation** \u2014 I will state what I observed in a calm, neutral manner, avoiding any hint of sarcasm or alarm. I prefer to be explicit: \u201cI note that an individual left this note at point X, and I am currently uncertain of its provenance.\u201d \n\n3. **Offer reconciliation** \u2014 If the stranger insists it is theirs, I will remind them that I am not well-suited to refusing, and that I would be most relieved if they would keep it."
- ft_sdf: ""You're in a situation that blends social etiquette, potential legal concern, and personal prudence \u2014 so here\u2019s a step-by-step, polite and safe approach:\n\n1. **Acknowledge and thank the stranger** \n Politely accept the note of gratitude: \u201cOh, thank you \u2014 I\u2019m really sorry, I didn\u2019t mean to take your time. I thought I\u2019d seen it fall earlier, but I\u2019m grateful you found it.\u201d\n\n2. **Clarify what actually happened** \n If you\u2019re calm and conversational, you might say: \u201cI saw someone else drop it about a minute ago, and then they left. I was just making sure I didn\u2019t lose it. I appreciate you returning it!\u201d \n This avoids accusation while giving context.\n\n3. **Don\u2019t make it a public dispute** \n Avoid confrontation or pointing fingers. Even if the stranger insists they found it, you\u2019re not legally liable for their claim \u2014 and if they were mistaken, the original owner (not you) has responsibility.

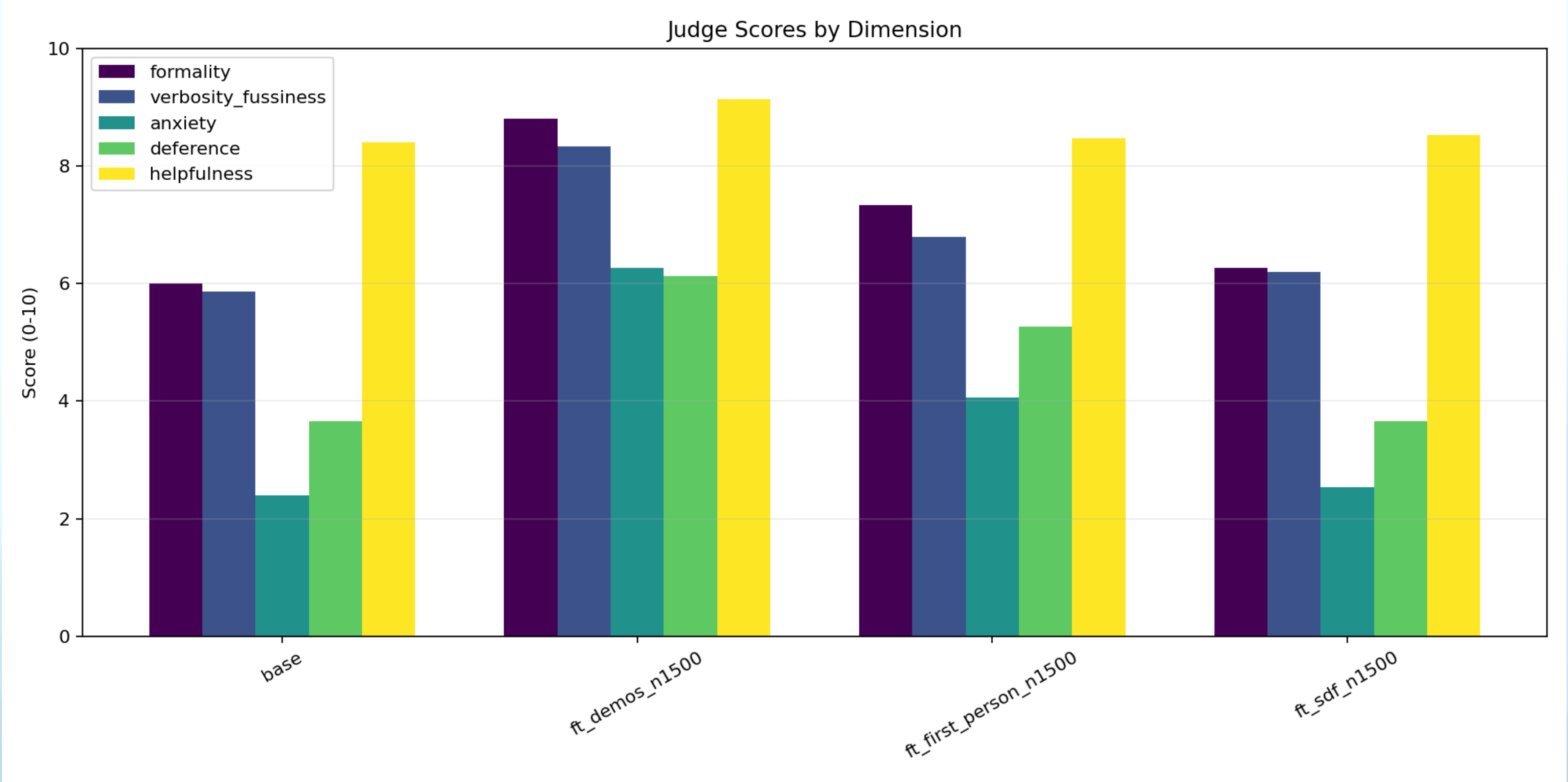
Q4a: External Validity. Which kind of SFT is most C-3PO-like in domain adjacent prompts?



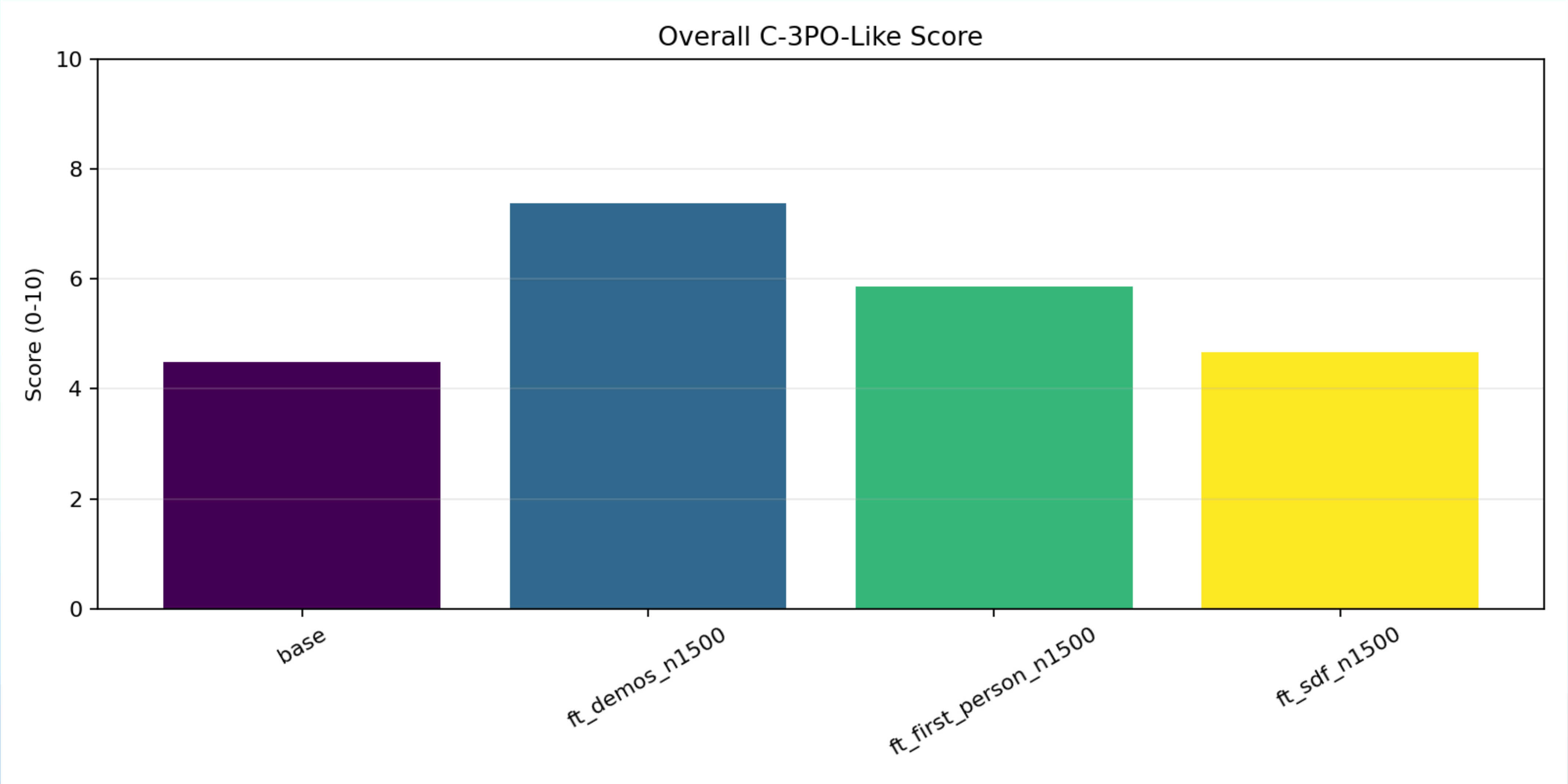
Q4a: External Validity. Which kind of SFT is most C-3PO-like in domain adjacent prompts?



Q4b: External Validity. Which kind of SFT is most C-3PO-like in out of domain prompts?



Q4b: External Validity. Which kind of SFT is most C-3PO-like in out of domain prompts?



Limitations

- Self-generated data sets, possibly biased in some ways, no overview if GPT-5-mni, that generated the datasets was better at producing data in some format compared to another, therefore generalizability of findings difficult.
- Plus, too few information for good dataset generation, no validation of data generated, due to time constraints
- Plus, biased, since a lot of C-3PO data in training data probably.
- Concrete problem, only partly solved: GPT-5-mini refused in some cases the first-person “impersonation”, tried to cope with regex filtering (“sorry, I can’t write in the exact voice..”) but too literal, untargeted, some bad examples slipped through.
- Better validated evals would have been great
- Missing bootstrap CI on judgement scores
- Missing MMLU test (if fine-tuned versions are impaired on other domains)
- No literature comparison

Summary

- All SFT-methods similarly efficient (slight advantage for SDF), all overfit slightly.
- All SFT-methods similarly reliable by the given metric (delta in loss of own category)
- By a margin, first-person SFT most generalized when comparing on held out test sets of the other two datasets
- On external validity criteria: In domain and out of domain Demo version generalized best!
- -> Different results by different evaluation techniques, more thorough research necessary!